Exact attained five-cycle thresholds for primal-first Chambolle–Pock dynamics

Abstract

We determine the least-period-five existence thresholds for the literal primal-first Chambolle–Pock map at extrapolation parameters 1/8 and 1. The thresholds are algebraic numbers of degree twelve, approximately 3.114241184076725 and 1.730020332057708, respectively. Each is specified by an integer polynomial and a certified rational isolating interval. Exact algebraic endpoint witnesses and strictly positive dual certificates prove that the feasible products are precisely the closed upper rays starting at these numbers. Necessity applies to arbitrary finite-dimensional real spaces, proper closed convex proximal functions, and linear couplings, using the actual operator norm. Endpoint constructions use convex-hull indicators in $\mathbb{R}^8$ and have a saddle pair. We include the complete eight-inequality Fourier–SOC reduction, the realization proof, the active-face matrices, and portable exact certificate checkers. This is a finite-cycle existence result; no global convergence or originality claim is made.

Keywords: Chambolle–Pock iteration; period-five thresholds; algebraic endpoints; Fourier–SOC criterion; exact dual certificates; endpoint attainment

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1 The map and the exact thresholds

Let X, Y be finite-dimensional real Hilbert spaces, let $L : X \to Y$ be linear, and let f, h be proper closed convex functions. For $\tau, \sigma > 0$, consider exactly

$$\begin{aligned} x_{i+1} &= \operatorname{prox}_{\tau f}(x_i - \tau L^* y_i), \\ y_{i+1} &= \operatorname{prox}_{\sigma h}\big(y_i + \sigma L((1+\theta)x_{i+1} - \theta x_i)\big). \end{aligned} \tag{1}$$

Here $\operatorname{prox}_{af}(z) = \arg\min_x\{f(x) + \|x - z\|^2/(2a)\}$. The state is the literal pair (x_i, y_i); no history variable is initialized. Write $r = \tau\sigma\|L\|^2$ for the actual product. A saddle pair satisfies $-L^* y_* \in \partial f(x_*)$ and $Lx_* \in \partial h(y_*)$.

Theorem 1. *For either $\theta = 1/8$ or $\theta = 1$, a nonconstant five-periodic paired orbit of (1) exists for some such convex problem with actual product r if and only if $r \geq \rho_\theta$. The same equivalence holds if a saddle pair is required. Every nonconstant five-periodic orbit has least period five. The number ρ_θ is the unique root of the corresponding polynomial below in $(L_\theta, L_\theta + 10^{-60})$.*

The exact rational left endpoints are

$$\begin{aligned} L_{1/8} &= 3 + \frac{1142411840767251140140433445 84}{10^{30}} \\ &\quad + \frac{3160745552825176704862855936 97}{10^{60}}, \\ L_1 &= 1 + \frac{7300203320577080235566078494 43}{10^{30}} \\ &\quad + \frac{5700104695374645915199261238 76}{10^{60}}. \end{aligned} \tag{2}$$

The decimal approximations in the abstract are only orientation; these rational intervals and the polynomial identities are the definitions and certificates.

The primitive irreducible integer polynomials are

$$\begin{aligned} F_{1/8}(r) &= 3413933326183502470 41r^{12} \\ &\quad - 4422046585153148331336r^{11} \\ &\quad + 1924725425959826853768 0r^{10} \\ &\quad - 1444694228600444702208 0r^{9} \\ &\quad - 1333232571672474149632 00r^{8} \\ &\quad + 3858064526976719794176 00r^{7} \\ &\quad - 1330443279689661939712 00r^{6} \\ &\quad - 9155145361232465231872 00r^{5} \\ &\quad + 1451090624848745988096 000r^{4} \\ &\quad - 5732036395161301811200 00r^{3} \\ &\quad - 2599199873287134904320 00r^{2} \\ &\quad + 1415050733999567667200 00r \\ &\quad + 3743779436936298496 0000, \end{aligned} \tag{3}$$

$$\begin{aligned} F_1(r) &= 42050000r^{12} - 354745000r^{11} \\ &\quad + 1183664725r^{10} - 1809261130r^{9} \\ &\quad + 649262107r^{8} + 1822783864r^{7} \\ &\quad - 2504247241r^{6} + 620333130r^{5} \\ &\quad + 893970565r^{4} - 640469600r^{3} \\ &\quad + 65345800r^{2} + 29388000r + 1922000. \end{aligned} \tag{4}$$

2 The complete period-five conic criterion

Put $\phi_k = 2\pi k/5$ for $k = 1, 2$. For $j = 1, 2, 3, 4$, define

$$\begin{aligned} D_{kj} &= \cos((j+1)\phi_k) - \cos\phi_k - \cos(j\phi_k) + 1, \\ Z_{kj} &= e^{\mathrm{i}\phi_k}(e^{\mathrm{i}j\phi_k} - 1), \\ W_{kj} &= ((1+\theta) - \theta e^{\mathrm{i}\phi_k})(e^{\mathrm{i}j\phi_k} - 1). \end{aligned} \tag{5}$$

The variables A_k, B_k, c_k, d_k satisfy

$$A_k, B_k \geq 0, \quad c_k^2 + d_k^2 \leq A_k B_k, \quad A_1 + A_2 = 1, \tag{6}$$

and all eight inequalities

$$\begin{aligned} P_j &= \sum_{k=1}^{2} \left(\frac{D_{kj}}{r} A_k - (\operatorname{Re} Z_{kj}) c_k - (\operatorname{Im} Z_{kj}) d_k \right) \leq 0, \\ H_j &= \sum_{k=1}^{2} (D_{kj} B_k + (\operatorname{Re} W_{kj}) c_k - (\operatorname{Im} W_{kj}) d_k) \leq 0. \end{aligned} \tag{7}$$

Lemma 1. *For $r > 0$, existence of a nonconstant five-periodic orbit of (1) is equivalent to feasibility of (6)–(7). Every feasible point is realized in $X = Y = \mathbb{R}^8$ with $L = I$, $\tau = r$, $\sigma = 1$, convex-hull indicators, and saddle pair $(0, 0)$.*

Proof. All indices are modulo five. Proximal optimality on an actual orbit gives

$$\begin{aligned} a_i &= (x_{i-1} - x_i)/\tau - L^* y_{i-1} \in \partial f(x_i), \\ b_i &= (y_{i-1} - y_i)/\sigma + L((1+\theta)x_i - \theta x_{i-1}) \in \partial h(y_i). \end{aligned} \tag{8}$$

Summing the subgradient inequalities to the displaced vertices cancels the function values. For each $j = 1, 2, 3, 4$ this yields

$$\sum_i \langle a_i, x_{i+j} - x_i \rangle \leq 0, \qquad \sum_i \langle b_i, y_{i+j} - y_i \rangle \leq 0.$$

Every lag is necessary; retaining only pairwise monotonicity would not suffice.

Center and scale the sequences by $U_i = (x_i - \bar{x})/\sqrt{\tau}$, $V_i = (y_i - \bar{y})/\sqrt{\sigma}$, and put $K = \sqrt{\tau\sigma} L$. Use the temporal inner product $5^{-1} \sum_i \langle \cdot, \cdot \rangle$. Let T be forward shift and $S = T^{-1}$. The last inequalities become

$$\begin{aligned} \langle (S - I)U - K^* S V, (T^j - I)U \rangle \leq 0, \\ \langle (S - I)V + K((1+\theta)I - \theta S)U, (T^j - I)V \rangle \leq 0. \end{aligned} \tag{9}$$

The centered temporal space splits orthogonally into modes $k = 1, 2$, on which $T = \cos\phi_k I + \sin\phi_k J_k$, with $J_k^* = -J_k$, $J_k^2 = -I$. The temporal lift of K intertwines these structures and has norm exactly $\sqrt{r}$. Set

$$A_k = r\|U_k\|^2, \quad B_k = \|V_k\|^2, \quad c_k = \langle K U_k, V_k \rangle, \quad d_k = \langle K J_k U_k, V_k \rangle.$$

The two vectors KU_k, KJ_kU_k are orthogonal with equal norms. Bessel's inequality and the actual norm bound give $c_k^2 + d_k^2 \le A_k B_k$, including when $KU_k = 0$. Expanding (9) on each mode gives precisely (7): the self coefficient is D_{kj}, the primal cross multiplier is Z_{kj}, and the dual multiplier after taking the adjoint is $\overline{W_{kj}}$.

If $U = 0$, the dual inequality at $j = 4$ gives $\|(S - I)V\|^2 \le 0$, so $V = 0$. Thus a nonconstant orbit has $\sum A_k > 0$. Dividing all four sets of variables by this sum yields (6), without changing the homogeneous inequalities. This proves necessity for arbitrary spatial dimensions and arbitrary convex functions.

For sufficiency take two orthogonal planes per mode, and let R_k rotate both planes through ϕ_k. For $A_k > 0$ define

$$\begin{aligned} p_k &= (\sqrt{A_k}, 0\,;\, 0, 0), \\ q_k &= \left(c_k/\sqrt{A_k}, d_k/\sqrt{A_k}\,;\, \sqrt{B_k - (c_k^2 + d_k^2)/A_k}, 0 \right). \end{aligned} \tag{10}$$

If $A_k = 0$, use $p_k = 0$ and $q_k = (0, 0\,;\, \sqrt{B_k}, 0)$; the cone forces $c_k = d_k = 0$. Thus every square root and every denominator is justified. Let $R = R_1 \oplus R_2$, $x_i = R^i(p_1 \oplus p_2)$, and $y_i = R^i(q_1 \oplus q_2)$. Take

$$C = \operatorname{conv}\{x_0, \dots, x_4\}, \quad E = \operatorname{conv}\{y_0, \dots, y_4\}, \quad f = \iota_C,\ h = \iota_E,\ L = I_8,\ \tau = r,\ \sigma = 1.$$

Define a_i, b_i by (8) with these choices. Their vertex tests are

$$\langle a_0, x_j - x_0 \rangle = P_j, \qquad \langle b_0, y_j - y_0 \rangle = H_j.$$

Rotation equivariance gives the same tests at every vertex. The eight inequalities therefore imply all forty off-diagonal normal comparisons. By linearity they hold on the whole hulls, giving $a_i \in N_C(x_i)$ and $b_i \in N_E(y_i)$. These are exactly the two proximal updates. Single-valuedness implies that the actual run initialized at (x_0, y_0) is this orbit.

Since $\sum_{i=0}^{4} R^i = 0$, both hulls contain zero, and the zero vectors belong to their normal cones there. Hence $(0, 0)$ is a saddle pair. Normalization makes the primal orbit nonconstant. Its least period divides five and is not one, so is five. This proves sufficiency and the lemma. ∎

3 The active face and its two quadratic forms

Write $z = \sqrt{5} > 0$,

$$C_1 = (z - 1)/4, \quad C_2 = -(z + 1)/4, \quad s_k = \sin\phi_k > 0, \quad s_k^2 = 1 - C_k^2.$$

Change variables to $X_k = A_k/r$, $d_k = s_k e_k$, in the order

$$w = (X_1, X_2, B_1, B_2, c_1, c_2, e_1, e_2)^T.$$

The cones become $q_k(w) = rX_kB_k - c_k^2 - s_k^2e_k^2 \geq 0$. To specify the rows algebraically over $\mathbb{Q}(z)$, use Chebyshev polynomials T_j, U_j and put

$$\begin{aligned} D_{kj} &= T_{j+1}(C_k) - C_k - T_j(C_k) + 1, \\ z_{kj}^R &= T_{j+1}(C_k) - C_k, \qquad\qquad z_{kj}^I = U_j(C_k) - 1, \\ w_{kj}^R &= (1+\theta)(T_j(C_k) - 1) - \theta z_{kj}^R, \\ w_{kj}^I &= (1+\theta)U_{j-1}(C_k) - \theta z_{kj}^I. \end{aligned} \tag{11}$$

The P_j row has coefficients D_{kj} on X_k, $-z_{kj}^R$ on c_k, and $-s_k^2 z_{kj}^I$ on e_k. The H_j row has coefficients D_{kj} on B_k, w_{kj}^R on c_k, and $-s_k^2 w_{kj}^I$ on e_k. Unmentioned coefficients are zero.

Let M contain the five rows P_1, P_3, H_1, H_2, H_3, in that order. These rows are independent. Choosing the free coordinates (c_2, e_1, e_2) gives $w = Nv$, with the following complete matrices:

$$N_{1/8} = \begin{pmatrix} 0 & (-35+z)/32 & 1/16 \\ 0 & 1/16 & (-35-z)/32 \\ 9/8 & (-173+9z)/128 & (225+45z)/128 \\ 9/8 & 0 & (13+9z)/32 \\ 1 & (-25+5z)/16 & (25+5z)/16 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{12}$$

$$N_1 = \begin{pmatrix} 0 & (-7+z)/4 & 1/2 \\ 0 & 1/2 & (-7-z)/4 \\ 2 & -6+z & (15+3z)/2 \\ 2 & 0 & (3+z)/2 \\ 1 & (-15+3z)/4 & (15+3z)/4 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{13}$$

Here N_{X_k} denotes the row associated with X_k, and similarly for the other coordinates. Define the symmetric 3×3 matrices

$$Q_k(r) = \frac{r}{2}(N_{X_k}^T N_{B_k} + N_{B_k}^T N_{X_k}) - N_{c_k}^T N_{c_k} - s_k^2 N_{e_k}^T N_{e_k}. \tag{14}$$

Thus $q_k(Nv) = v^T Q_k(r) v$. The determinant pencil is

$$\mathcal{D}(r,u) = \det(Q_1(r) + uQ_2(r)) = a(r)u^3 + b(r)u^2 + c(r)u + d(r). \tag{15}$$

Its discriminant in u is a nonzero constant multiple of the corresponding F_θ in (3) or (4). Equations (12)–(15) specify the complete Gram pencil without large expanded rational tables. The source package also supplies its expanded coefficients and the scalar discriminant factors as readable exact data.

4 Algebraic endpoint witnesses and branch selection

Fix one of the two values of θ, and write $F = F_\theta$, $\rho = \rho_\theta$. Let rem_F denote entrywise polynomial remainder in r modulo F over $\mathbb{Q}(\sqrt{5})$. All expressions below are reduced first and evaluated at $r = \rho$ afterward. Irreducibility over $\mathbb{Q}(\sqrt{5})$ is not assumed: polynomial remainder identities are valid at the selected root regardless.

Define

$$\begin{gathered} n = \mathrm{rem}_F(9ad - bc), \qquad \eta = \mathrm{rem}_F(2(b^2 - 3ac)), \\ G = \mathrm{rem}_F(\eta Q_1 + nQ_2), \\ v = -\mathrm{rem}_F(\text{first column of } \operatorname{adj} G), \qquad w = \mathrm{rem}_F(Nv). \end{gathered} \tag{16}$$

"First column" means mathematical column one, or array index zero. This single convention is used throughout. No optimization or numerical root choice is hidden in (16).

The exact certificates establish

$$\eta > 0, \quad n > 0, \quad \eta^3 \mathcal{D}(\rho, n/\eta) = 0, \quad \eta^2 \partial_u \mathcal{D}(\rho, n/\eta) = 0. \tag{17}$$

Thus the denominator is nonzero and the selected repeated-root branch $u = n/\eta$ is positive. Discriminant vanishing alone would not select this branch. Further exact identities and strict signs are

$$\begin{gathered} Mw = 0, \qquad Gv = 0, \qquad v^T Q_1(\rho) v = v^T Q_2(\rho) v = 0, \\ X_1, X_2, B_1, B_2 > 0, \qquad P_2(w), P_4(w), H_4(w) < 0. \end{gathered} \tag{18}$$

In particular $w \neq 0$, so the chosen adjugate column is nonzero. The five remaining interpolation expressions vanish by $Mw = 0$. All signs and identities here are certified by the exact procedure in Section 7, not by the decimal approximations.

Set $Z = \rho(X_1 + X_2) > 0$, and define

$$\widehat{A}_k = \rho X_k / Z, \quad \widehat{B}_k = B_k / Z, \quad \widehat{c}_k = c_k / Z, \quad \widehat{d}_k = s_k e_k / Z. \tag{19}$$

Then $\widehat{A}_1 + \widehat{A}_2 = 1$, all eight inequalities hold, and $\widehat{A}_k \widehat{B}_k = \widehat{c}_k^2 + \widehat{d}_k^2$. All divisions are by strictly positive numbers. Applying (10) to these hatted variables gives fully specified algebraic endpoint coordinates. The final square root in each q_k is zero because the cone determinant vanishes. Both primal modes are present. Lemma 1 supplies the actual proximal orbit, the zero saddle pair, and least period five. This proves attainment.

5 Positive dual weights and exclusion of every lower product

The active face is used only to manufacture a certificate. The exclusion argument in this section imposes no active-face equalities on a hypothetical orbit.

Put $f_1 = \eta$, $f_2 = n$, and define the eight-coordinate vector g at the unnormalized endpoint witness by

$$\begin{aligned} g_{X_k} &= f_k \rho B_k, \quad g_{B_k} = f_k \rho X_k, \\ g_{c_k} &= -2 f_k c_k, \quad g_{e_k} = -2 f_k s_k^2 e_k. \end{aligned} \tag{20}$$

Let $J = (1, 2, 3, 4, 5)$, using mathematical column indices of M. The square matrix $M_{[:,J]}$ is nonsingular, and define

$$\ell = (M_{[:,J]}^T)^{-1} g_J = (\lambda_1, \lambda_3, \mu_1, \mu_2, \mu_3)^T. \tag{21}$$

Exact reduction verifies *all eight* equations $M^T \ell = g$, rather than only the five used to solve (21). Rational interval signs prove that all five weights are strictly positive. Set $\lambda_2 = \lambda_4 = \mu_4 = 0$.

For arbitrary feasible SOC variables at product $r > 0$, their nonnegative weighted interpolation sum is

$$\sum_j (\lambda_j P_j + \mu_j H_j) = \sum_{k=1}^{2} (\alpha_k(r) A_k + \beta_k B_k + \gamma_k c_k + \delta_k d_k) \le 0. \tag{22}$$

At $r = \rho$, the coefficient identities $M^T \ell = g$ give

$$\begin{aligned} \alpha_k(\rho) &= f_k B_k > 0, \quad \beta_k = f_k \rho X_k > 0, \\ \gamma_k &= -2 f_k c_k, \qquad \delta_k = -2 f_k s_k e_k. \end{aligned} \tag{23}$$

The coordinates on the right are the fixed endpoint witness, not the arbitrary variables in (22). Saturation of its cones yields

$$4 \alpha_k(\rho) \beta_k - \gamma_k^2 - \delta_k^2 = 0.$$

Equivalently the Hermitian matrices

$$H_k = \begin{pmatrix} \alpha_k(\rho) & (\gamma_k + \mathrm{i}\delta_k)/2 \\ (\gamma_k - \mathrm{i}\delta_k)/2 & \beta_k \end{pmatrix} \tag{24}$$

are positive semidefinite of rank one. These matrices explicitly describe the endpoint null directions; their kernels are generated by

$$\left(-\frac{\gamma_k + \mathrm{i}\delta_k}{2\alpha_k(\rho)}, 1 \right)^T.$$

In the real cone variables, equality in a nonzero mode occurs exactly on the ray through

$$(A, B, c, d) = (\rho X_k, B_k, c_k, s_k e_k). \tag{25}$$

Indeed Cauchy–Schwarz forces (c, d) opposite to (γ_k, δ_k) with length $\sqrt{AB}$, and equality in the remaining scalar arithmetic–geometric mean inequality forces $\alpha_k A = \beta_k B$. Positivity

of both diagonal entries excludes any nonzero equality point with $A = 0$ or $B = 0$. This proves the stated null-cone description, including the origin.

Only the A coefficient depends on the test product, and

$$\alpha_k(r) = \frac{\rho}{r}\alpha_k(\rho).$$

For every $0 < r < \rho$ we therefore have

$$4\alpha_k(r)\beta_k - \gamma_k^2 - \delta_k^2 = 4\alpha_k(\rho)\beta_k(\rho/r - 1) > 0. \tag{26}$$

For any cone point, Cauchy–Schwarz gives

$$\alpha_k(r)A + \beta_k B + \gamma_k c + \delta_k d \geq \alpha_k(r)A + \beta_k B - \sqrt{\gamma_k^2 + \delta_k^2}\sqrt{AB}.$$

By (26) this is strictly positive whenever $A + B > 0$: for $AB > 0$ compare the cross coefficient with $2\sqrt{\alpha_k(r)\beta_k}$, and for $AB = 0$ use the positive diagonal terms. Since $A_1 + A_2 = 1$, at least one mode is nonzero. Thus (22) is strictly positive, a contradiction. Lemma 1 makes this exclusion universal across all convex problems and dimensions, with or without saddle points.

If $r = 0$, then $L = 0$. The two proximal iterations decouple. Summing the predecessor subgradient inequalities on each periodic sequence gives $\sum_i \|x_i - x_{i-1}\|^2/\tau \leq 0$ and its dual analogue. Both sequences are constant. This treats the zero product without dividing by it.

6 Every larger actual norm product

Fix any $r' > \rho$. Keep the endpoint realization, steps $\tau = \rho$, $\sigma = 1$, and orbit. Append one unused orthogonal coordinate to each space, constrain both extra coordinates to zero by indicator functions, and take the coupling

$$L' = I_8 \oplus \sqrt{r'/\rho}.$$

The actual norm is $\sqrt{r'/\rho}$, so $\tau\sigma\|L'\|^2 = r'$ exactly. The extra coordinates remain zero, and the original orbit and zero saddle pair are unchanged. This establishes existence for *every* larger product, without any unsupported monotonicity inference about the signed coefficients in (7). If desired, one can use an unused coordinate already inside $\mathbb{R}^8$, because each saturated endpoint mode occupies only its first plane. Together with attainment and exclusion, this proves Theorem 1.

7 Exact verification and reproducible certificates

The accompanying source package contains two portable verification routes using Python, SymPy, and the standard library. The file `verify_exact.py` uses

`derive.py` to reconstruct the Chebyshev rows and their nullspace. The independent file `check_exact.py` imports no author derivation or stored certificate: it reconstructs the rows by real-pair multiplication representing complex rotations, solves the five active equations directly, and compares the result with both printed N matrices. Both routes regenerate the determinant pencil and discriminant. There is no serialized exploration object and no numerical feasibility oracle.

The independent route checks the following finite exact statements for each parameter:

1. The printed polynomial has integer content one, degree twelve, and is irreducible over $\mathbb{Q}$, using exact polynomial irreducibility. It is therefore the minimal polynomial of the selected endpoint.

2. A Sturm chain starts with F, F' and continues by negative exact Euclidean remainders until a nonzero constant. Neither interval endpoint is a root; the difference of sign variations is one. Thus the width-10^{-60} rational interval isolates exactly one real root.

3. The first five columns of M have nonzero determinant. Direct solution gives (12) or (13), and $MN = 0$. The discriminant divided by F is a nonzero constant.

4. The two cleared-denominator repeated-root identities in (17), the identities in (18), and both cone equalities have zero polynomial remainder modulo F. The positive signs of $n, \eta, X_1, X_2, B_1, B_2$ and negative signs of the other three rows are certified.

5. The dual solution satisfies all eight equations, all five weights are positive, both endpoint diagonal coefficients are positive, and both Hermitian determinants vanish exactly.

Every strict sign is established by rational Horner interval arithmetic. In addition to (2), the checker encloses $z = \sqrt{5}$ between the exact terminating decimals obtained by concatenating these fractional lines:

$$2.2360679774997896964091736687312762354406183596115257242708 97$$

and the number 10^{-60} larger. Rational squaring verifies that the lower endpoint squared is below five and the upper endpoint squared is above five. For a reduced polynomial $p(r) = \sum_j (a_j + b_j z) r^j$, interval Horner evaluation encloses its exact value at ρ. A positive sign is accepted only when the entire rational enclosure lies strictly above zero. Consequently every denominator, orientation, unused inequality, and positive weight used in the proof is checked with no floating-point tolerance.

The checker writes `exact_certificate.json`, a human-readable exact data file containing both polynomials, isolating intervals, Sturm variations, full row and nullspace matrices, quadratic forms, determinant pencils, discriminant factors, cofactor vectors,

endpoint witnesses, and dual weights. The file is an inspectable record, not an input trusted by either checker. The identities are rebuilt from the mathematical definitions each time. Run `python check_exact.py` and `python verify_exact.py`; both must complete without failed assertions. Run `bash build.sh` to rebuild this manuscript.

The complete SOC necessity and realization proofs above explain why these finite algebraic checks imply the theorem. Polynomial elimination by itself would neither exclude other feasible faces nor establish actual proximal dynamics. The strictly positive universal dual certificate supplies the former, and the explicit hull-normal construction supplies the latter. The result asserts only the sharp existence boundary for least-period-five orbits at the two specified extrapolation parameters. It makes no statement of convergence below that boundary or of historical priority.